4.3.1 An unloaded potential divider

In an unloaded potential divider, the series resistors carry the same current. The output's fraction of supply voltage equals the output resistor's fraction of total resistance. Here the output is across the first resistor. A connected load is in parallel with this resistor and changes the division ratio. Find their equivalent resistance, then recalculate the output voltage.

$$V_{\text{out}} = \frac{R_1}{R_1 + R_2}\varepsilon$$

Unloaded potential divider. Output is across R1 in the source diagram. R1, R2: Ω; ε, V_{out}: V. A connected load changes the effective resistance.

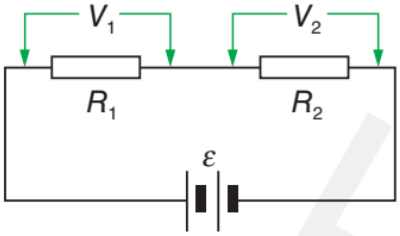


Fig 3.1d *Resistors R_1 and R_2 connected in series to a battery.*

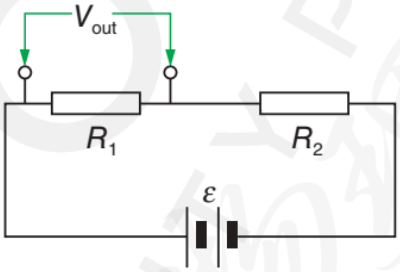


Fig 3.1e *The output voltage depends on the resistance ratio.*

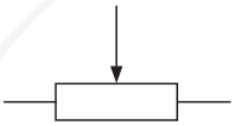


Fig 3.1f *Circuit symbol of potential divider.*

The output terminals are across R1. This is why R1, rather than R2, appears in the numerator of the displayed divider formula.

4.3.2 Connecting an ammeter

An ammeter is connected in series so that the measured branch current passes through it. An ideal ammeter has zero resistance. A real meter disturbs the current less when its resistance is low. Connecting it across a power supply can create a near short circuit.

4.3.3 Connecting a voltmeter and loading effects

A voltmeter is connected in parallel across the two points being measured. Ideally its resistance is infinite and it draws no current. A real voltmeter draws some current and can change the original voltage distribution. Higher meter resistance generally causes less disturbance, especially when measuring high-resistance circuits.

4.3.4 Internal resistance and terminal voltage

For a discharging cell, e.m.f. equals external terminal voltage plus the drop across internal resistance. Greater current increases this internal drop and lowers terminal voltage. At zero current, terminal voltage equals e.m.f. During charging, current reverses, so the discharge subtraction relation cannot be applied without changing the sign convention.

$$\varepsilon = V + Ir$$

Discharging cell: ε: e.m.f. (V); V: terminal voltage (V); I: supplied current (A); r: internal resistance (Ω). During charging the terminal-voltage relation changes sign.